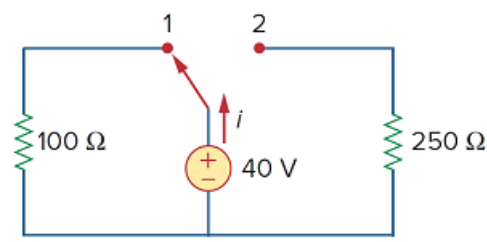


Problem

(a) Calculate current i in Fig. 2.68 when the switch is in position 1.

(b) Find the current when the switch is in position 2.

Figure 2.68:



Step-by-step solution

Step 1 of 3

The circuit is shown in Figure 1.

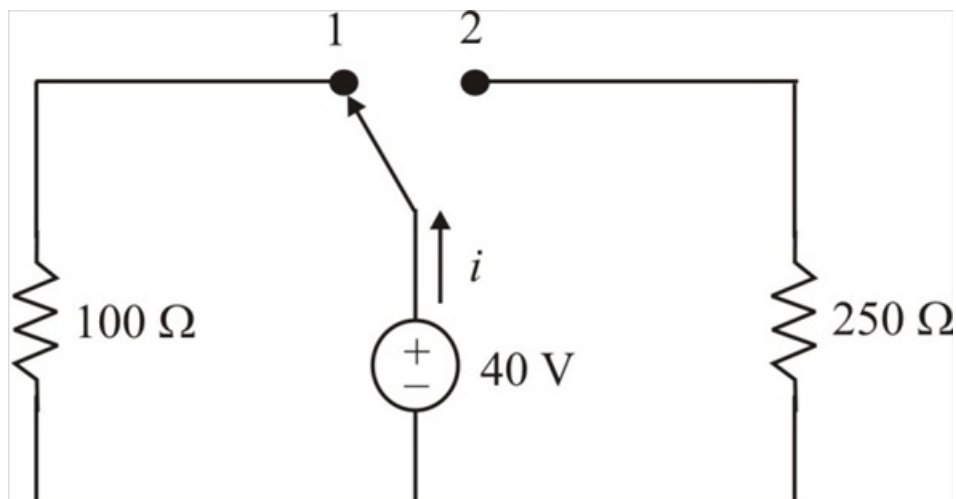


Figure 1

Step 2 of 3

(a)

When the switch is in position 1:

Applying Ohm's law, calculate the current i .

$$\begin{aligned} i &= \frac{v}{R} \\ &= \frac{40}{100} \\ &= 400 \text{ mA} \end{aligned}$$

Therefore, the current i is 400 mA.

Step 3 of 3

(b)

When the switch is in position 2:

Apply Ohm's law, calculate the current i .

$$\begin{aligned} i &= \frac{v}{R} \\ &= \frac{40}{250} \\ &= 160 \text{ mA} \end{aligned}$$

Therefore, the current i is 160 mA.